SLIC Index V3

Methodology Technical Paper

Smart and Liveable Cities Index

157 Published Cities · 5 Pillars · 36 Indicators

Non Arkara

Associate Professor Poon Thiengburanathum

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This document is the complete technical specification for the SLIC Index scoring system. It details every formula, every threshold, and every data source used to produce the published rankings. The methodology is designed for full reproducibility: any researcher with access to the listed sources can independently verify every score in the index.

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1. Executive Summary

The Smart and Liveable Cities Index (SLIC) is a transparent, open-source city ranking system that measures quality of life for people building their lives in cities around the world. Unlike existing indices that optimize for expat comfort, investor sentiment, or editorial taste, SLIC focuses on what matters to residents: disposable income after rent, safety, healthcare access, tolerance, creative opportunity, and the ability to build a meaningful life.

SLIC V3 ranks 157 published cities across 5 pillars and 36 indicators. Every score is computed using fixed absolute benchmarks—adding city #501 never changes the score for cities #1–500. There is no black box: every raw value traces through a declared piecewise linear mapping to a 0–100 normalized score, aggregated via the Adjusted Mazziotta–Pareto Index (AMPI) which penalizes imbalance. A city scoring 90/20 across indicators is penalized more than a city scoring 55/55.

The Five Pillars

Pillar	Weight	Focus	# Indicators
Growth	25%	Economic dynamism, entrepreneurship, innovation	7
Viability	22%	Safety, environment, healthcare, real income	8
Capability	18%	Transit, digital, education, walkability	7
Community	15%	Tolerance, equality, trust, belonging	7
Creative	20%	Culture, cuisine, nightlife, events, arts	7

Key Design Choices

Absolute scoring. Every indicator is mapped to 0–100 against fixed, published thresholds. This means scores are comparable across time and editions without recalibration. A city that scores 72 today scores 72 next year if its raw data hasn't changed.

No imputation. SLIC does not fabricate data for missing indicators. Instead, missing indicators are excluded from aggregation and the resulting data gap is disclosed via a coverage grade (A, B, or C). Grade B carries a 5-point penalty; Grade C carries a 15-point penalty.

Open source. The complete scoring code, data pipeline, and this methodology document are published under open license. Any researcher can reproduce every score.

2. Scoring Framework

2.1 Piecewise Linear Normalization

All indicators are normalized to a 0–100 scale using piecewise linear interpolation between fixed anchor points. Each indicator has 5 anchors mapping raw values to scores of 0, 25, 50, 75, and 100. Values beyond the outer anchors are clamped.

$$score = score_i + (score_{i+1} - score_i) \times (raw - raw_i) / (raw_{i+1} - raw_i)$$

For “lower is better” indicators (e.g., homicide rate, PM2.5, housing burden), the anchor mapping is reversed: the lowest raw value maps to 100 and the highest to 0.

2.2 Adjusted Mazziotta–Pareto Index (AMPI)

Pillar scores and the overall SLIC score are computed using AMPI, which penalizes imbalance across component scores. The formula:

$$mu = (1/n) \times SUM(s_i) \qquad sigma = sqrt((1/n) \times SUM((s_i - mu)^2)) \qquad cv = sigma / mu$$

$$AMPI = mu - (sigma \times cv) = mu - sigma^2 / mu$$

This means a city scoring 90/20 across two indicators gets a lower AMPI than a city scoring 55/55, even though their arithmetic means are similar (55 vs. 55). The variance penalty ensures that cities cannot hide weaknesses behind a single strong dimension.

2.3 Overall SLIC Score

The overall SLIC score applies AMPI across the five pillar scores:

```
SLIC = AMPI(Growth, Viability, Capability, Community, Creative)
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All five pillars carry declared weights (Growth 25%, Viability 22%, Capability 18%, Community 15%, Creative 20%) which are applied as weighted inputs to the AMPI calculation. The final score is then adjusted by the coverage penalty.

2.4 Coverage Penalties

Grade	Condition	Penalty	Status
A	>= 6 of 7 indicators (or >= 5 of 5–6)	0 points	Full confidence
B	>= 4 indicators present	–5 points	Published with note
C	<= 3 indicators present	–15 points	Flagged provisional

The penalty is applied after AMPI computation: pillar_score = max(0, AMPI – coverage_penalty). No imputation is performed. Missing indicators are excluded from aggregation.

3. Pillar: Growth (Economic Dynamism)

Captures expansion, entrepreneurial ecosystems, innovation output, and civic freedom through fixed public thresholds. This pillar rewards cities that foster economic opportunity and entrepreneurial dynamism—not just GDP size, but the environment for building businesses and careers.

Indicators

#	Indicator	Unit	Direction	Data Source
G1	Real GDP growth rate (5yr avg)	% p.a.	Higher better	IMF WEO, OECD
G2	Startup density	Per 100k pop	Higher better	Crunchbase, Dealroom
G3	VC investment intensity	USD per capita (3yr)	Higher better	Crunchbase, PitchBook
G4	Ease of doing business	Score 0–100	Higher better	World Bank B-READY
G5	Civic freedom	Score 0–100	Higher better	Freedom House + V-Dem
G6	Patent applications	Per 100k pop (3yr)	Higher better	WIPO
G7	High-skill employment share	% workforce ISCO 1–3	Higher better	ILO, LinkedIn

Normalization Anchors

G1 — Real GDP growth (% p.a.)

Raw Value	Score
≤0	0
1.0	25
2.5	50
4.0	75
≥6.0	100

G2 — Startup density (per 100k)

Raw Value	Score
≤5	0
20	25
50	50
100	75
≥200	100

G3 — VC investment (USD/capita, 3yr avg)

Raw Value	Score
≤5	0
50	25

200	50
500	75
≥1500	100

G4 — Ease of doing business

Direct passthrough (0–100 raw score).

G5 — Civic freedom

Composite: 0.6 × Freedom House + 0.4 × V-Dem liberal democracy index, both rescaled 0–100.

G6 — Patents per 100k (3yr avg)

Raw Value	Score
≤2	0
15	25
40	50
80	75
≥150	100

G7 — High-skill employment (%)

Raw Value	Score
≤10	0
20	25
30	50
40	75
≥55	100

4. Pillar: Viability (Lived Sustainability)

Measures safety, environmental quality, healthcare access, and real disposable income after all essential costs. This pillar asks: can an ordinary resident live safely and healthily without financial distress? It captures purchasing power after rent (DI_PPP), not headline GDP.

Indicators

#	Indicator	Unit	Direction	Data Source
V1	Homicide rate	Per 100k pop	Lower better	UNODC
V2	Air quality (PM2.5)	$\mu g/m^3$	Lower better	WHO, IQAir
V3	Healthcare access & quality	HAQ Index 0–100	Higher better	IHME/Lancet
V4	PPP-adjusted disposable income	USD/month	Higher better	Numbeo + World Bank
V5	Housing burden	Rent % of gross income	Lower better	Numbeo, surveys
V6	Water & sanitation quality	% safely managed	Higher better	WHO/UNICEF JMP
V7	Peace & political stability	GPI inverted	Higher better	IEP GPI
V8	Birth rate (optimism signal)	Per 1000 pop	Higher better	UN Population Div

Normalization Anchors

V1 — Homicide rate (per 100k, lower is better)

Raw Value	Score
≥50	0
20	25
5	50
1.0	75
≤0.3	100

V2 — PM2.5 ($\mu g/m^3$, lower is better)

Raw Value	Score
≥80	0
50	25
25	50
10	75
≤5	100

V3 — Healthcare (HAQ 0–100)

Direct passthrough.

V4 — DI_PPP (USD/month after rent, utilities, transit, food)

Formula: DI_PPP = (gross_income × (1 – tax_rate) – rent – utilities – transit – internet – food) / ppp_factor

Raw Value	Score
≤0	0
200	15
500	35
1000	55
2000	75
≥4000	100

V5 — Housing burden (% of gross income, lower is better)

Raw Value	Score
≥70	0
50	25
30	50
20	75
≤10	100

V6 — Water & sanitation (%)

Raw Value	Score
≤50	0
70	25
85	50
95	75
≥99	100

V7 — Peace (GPI inverted)

GPI 1.0–4.0 → score = max(0, min(100, (4 – GPI) / 3 × 100))

V8 — Birth rate (per 1000)

Raw Value	Score
≤4	0
7	25
10	50
14	75
≥20	100

5. Pillar: Capability (Infrastructure & Systems)

How well does the city's built environment serve residents? This pillar measures transit coverage, digital infrastructure, education quality, walkability, cycling infrastructure, and renewable energy adoption.

Indicators

#	Indicator	Unit	Direction	Data Source
C1	Transit coverage	% pop within 500m	Higher better	GTFS, ITDP
C2	Internet speed	Mbps median	Higher better	Ookla
C3	Digital government	EGDI 0–100	Higher better	UN E-Gov Survey
C4	Education quality	PISA score	Higher better	OECD PISA, UNESCO
C5	Walkability	Walk Score 0–100	Higher better	Walk Score, OSM
C6	Cycling infrastructure	km/100k pop	Higher better	OSM CyclOSM
C7	Renewable energy share	% electricity	Higher better	CDP, IRENA

Normalization Anchors

C1 — Transit coverage (%)

Raw Value	Score
≤5	0
20	25
40	50
65	75
≥85	100

C2 — Internet speed (Mbps)

Raw Value	Score
≤5	0
25	25
75	50
150	75
≥300	100

C3 — Digital government (EGDI × 100)

Direct passthrough.

C4 — Education (PISA avg)

Tertiary fallback: ≤5%=0, 15%=25, 25%=50, 40%=75, ≥55%=100

Raw Value	Score
≤350	0
400	25
450	50
500	75
≥550	100

C5 — Walkability (0–100)

Direct passthrough.

C6 — Cycling (km per 100k)

Raw Value	Score
≤1	0
5	25
15	50
30	75
≥50	100

C7 — Renewables (%)

Raw Value	Score
≤2	0
15	25
35	50
60	75
≥90	100

6. Pillar: Community (Belonging & Tolerance)

Does the city welcome you regardless of who you are? This pillar measures inclusivity, social trust, gender equality, and income distribution. It penalizes cities where belonging is conditional on identity.

Indicators

#	Indicator	Unit	Direction	Data Source
O1	LGBTQ+ rights	Composite 0–100	Higher better	ILGA, Equaldex
O2	Religious freedom	Pew GRI inverted	Higher better	Pew Research
O3	Immigrant integration	Employment gap (pp)	Lower better	OECD
O4	Income inequality	Gini 0–1	Lower better	World Bank
O5	Social trust	% trusted	Higher better	WVS, Gallup
O6	Gender equality	GII inverted 0–100	Higher better	UNDP
O7	Weekend retention	Weekend/weekday ratio	Higher better	Mobile analytics

Normalization Anchors

O1 — LGBTQ+ rights (composite 0–100)

Built from: same-sex marriage (25pts), anti-discrimination (25pts), gender recognition (15pts), no criminalization (20pts), social acceptance (15pts).

O2 — Religious freedom (GRI inverted)

score = max(0, min(100, (10 – GRI) / 10 × 100))

O3 — Immigrant integration (pp gap, lower is better)

Raw Value	Score
≥30	0
20	25
10	50
5	75
≤1	100

O4 — Gini (lower is better)

Raw Value	Score
≥0.60	0
0.45	25
0.35	50
0.28	75
≤0.22	100

O5 — Social trust (%)

Raw Value	Score
≤5	0
15	25
30	50
50	75
≥70	100

O6 — Gender equality (GII inverted)

score = (1 – GII) × 100

O7 — Weekend retention ratio

Raw Value	Score
≤0.70	0
0.80	25
0.90	50
1.00	75
≥1.10	100

7. Pillar: Creative (Cultural Richness)

Is the city culturally alive? This pillar measures cultural venues, heritage access, culinary diversity, nightlife, arts funding, creative employment, and international events. It penalizes cultural deserts regardless of economic strength.

Indicators

#	Indicator	Unit	Direction	Data Source
R1	Cultural venues density	Per 100k pop	Higher better	OSM, Google Places
R2	UNESCO heritage proximity	Sites within 50km	Higher better	UNESCO + GIS
R3	Culinary diversity	Cuisine types/100k	Higher better	Google, Yelp
R4	Nightlife density	Bars+clubs/100k	Higher better	OSM, Google Places
R5	Arts & culture funding	USD PPP/capita	Higher better	Eurostat, ministries
R6	Creative industry employment	% workforce	Higher better	UNCTAD
R7	International events	Per million pop/yr	Higher better	ICCA, Eventbrite

Normalization Anchors

R1 — Cultural venues (per 100k)

Raw Value	Score
≤2	0
8	25
20	50
40	75
≥70	100

R2 — UNESCO sites within 50km

Raw Value	Score
0	0
1	25
3	50
6	75
≥10	100

R3 — Culinary diversity (per 100k)

Raw Value	Score
≤1	0
3	25
8	50

15	75
≥25	100

R4 — Nightlife (per 100k)

Raw Value	Score
≤3	0
10	25
25	50
50	75
≥80	100

R5 — Arts funding (USD PPP/capita)

Raw Value	Score
≤5	0
30	25
80	50
150	75
≥300	100

R6 — Creative employment (%)

Raw Value	Score
≤1	0
3	25
6	50
10	75
≥15	100

R7 — Events (per million pop/yr)

Raw Value	Score
≤2	0
10	25
25	50
50	75
≥100	100

8. Data Sources

The following table lists the primary data sources used across all 36 indicators. Each metric in the published ranking includes a source field and source URL for full traceability.

Source	Indicators
IMF WEO / OECD Regional	G1
Crunchbase / Dealroom / StartupBlink	G2, G3
World Bank B-READY	G4
Freedom House + V-Dem	G5
WIPO	G6
ILO / LinkedIn Talent Insights	G7
UNODC	V1
WHO / IQAir	V2
IHME / Lancet HAQ Index	V3
Numbeo + World Bank PPP	V4, V5
WHO/UNICEF JMP	V6
IEP Global Peace Index	V7
UN Population Division	V8
GTFS / ITDP	C1
Ookla Speedtest	C2
UN E-Government Survey	C3
OECD PISA / UNESCO	C4
Walk Score / OSM	C5
OSM CyclOSM	C6
CDP Cities / IRENA	C7
ILGA / Equaldex	O1
Pew Research Center	O2
OECD	O3
World Bank	O4
World Values Survey / Gallup	O5
UNDP	O6
Mobile analytics providers	O7
OSM / Google Places	R1, R3, R4
UNESCO World Heritage Centre	R2
Eurostat / national ministries	R5
UNCTAD	R6
ICCA / Eventbrite	R7

9. Coverage Grades & Missing Data

SLIC does not impute missing data. When an indicator is unavailable for a city, it is excluded from the aggregation. The resulting data coverage is disclosed via a letter grade:

Grade A: At least 6 of 7 indicators present (or 5 of 5–6). Full confidence. No penalty.

Grade B: At least 4 indicators present. Published with note. 5-point penalty applied after AMPI.

Grade C: 3 or fewer indicators present. 15-point penalty. City flagged as provisional.

The coverage grade is computed per pillar and overall. The penalty is floored at 0 (scores cannot go negative). This approach ensures transparency: users can see exactly how much data supports each city’s score.

Rules

1. No imputation. Missing indicators are excluded from aggregation.
2. Coverage penalty applied after AMPI computation, floored at 0.
3. Coverage grade is stored and displayed alongside every score for transparency.
4. Cities with Grade C are flagged as provisional and carry a disclaimer.

10. City Rankings Tabulation

The following table lists all 157 published cities in the SLIC Index V3, sorted by overall SLIC score. Pillar scores are shown on a 0–100 scale. Coverage grade indicates data completeness (A = best, C = provisional).

#	City	Country	SLIC	Grw	Via	Cap	Com	Cre	Grd
1	Kaohsiung	Taiwan	67.6	52.4	87.4	90.6	70.2	42	A
2	Taipei	Taiwan	66.9	40.5	85.1	89.7	68.1	58.7	A
3	Raleigh	United States	64	54.8	80.7	79.1	39.3	62	A
4	Busan	South Korea	63.7	47.2	92.3	85.8	40.5	50.6	A
4	Katowice	Poland	63.7	65.7	79.5	65.1	64.9	41.9	A
6	Fukuoka	Japan	62.4	50.7	92	67.4	46	52.3	A
7	Bangkok	Thailand	61.8	56.4	85.7	61.4	88.3	22.8	A
8	Lyon	France	61.6	42.2	84.3	63.9	73.7	50.1	A
8	Montreal	Canada	61.6	51.6	70.4	75.3	58.7	54.1	A
10	Santiago	Chile	61.3	54	78.6	83.8	43	44.6	A
11	Pittsburgh	United States	61.2	56.3	66.3	79.1	39.3	62	A
12	Kobe	Japan	61.1	49.5	94.6	67.4	42	47.3	A
13	Minneapolis	United States	60.6	52.4	68.1	79.1	39.3	62	A
14	Haifa	Israel	60.3	54.5	89.2	64.7	29.8	54.7	A
15	Suwon	South Korea	60.2	41.3	86.1	85.8	35.8	50.6	A
16	Ottawa	Canada	60	47	68.6	75.3	58.7	54.1	A
17	Valparaiso	Chile	59.8	55	72.7	83.8	40	44.6	A
18	Milan	Italy	59.5	72.2	76.4	66.9	45	29.4	A
19	Eindhoven	Netherlands	59.3	53.2	75.1	71.8	53.2	42.8	A
20	Graz	Austria	59.2	47.3	71.8	74.4	66.2	41.4	A
21	Braga	Portugal	59.1	52.5	84.6	68.8	41.7	43.8	A
21	Tel Aviv	Israel	59.1	46.2	91	64.7	32.8	54.7	A
23	Chicago	United States	58.9	42.5	71.7	79.1	39.3	62	A
24	Porto	Portugal	58.8	48.2	88.2	68.8	41.7	43.8	A
25	Tallinn	Estonia	58.7	47.5	59.3	58.4	58.1	72.8	A
25	Torun	Poland	58.7	62.3	79.5	65.1	49.9	31.9	A
27	Incheon	South Korea	58.6	41.3	86	85.8	25.5	50.6	A
27	Singapore	Singapore	58.6	35.2	74.8	94.1	11.3	73.3	A
29	Zurich	Switzerland	58.4	32.3	78.6	72.7	66.8	49.4	A
30	Copenhagen	Denmark	58.3	37.6	61.8	71.3	67.8	61.5	A
31	Khobar	Saudi Arabia	58	75	96.3	78.9	17.4	6.1	A
31	Sydney	Australia	58	22.4	94.1	87.1	40.5	49.7	A
33	Jeddah	Saudi Arabia	57.6	73.7	96.3	78.9	17.4	6.1	A
34	Brisbane	Australia	57.5	28.8	94.1	87.1	37.5	41.7	A
35	Vienna	Austria	57.2	37.5	73.6	74.4	66.2	41.4	A
36	Melbourne	Australia	56.9	30.8	82.9	87.1	39.5	46.7	A
36	Perth	Australia	56.9	28.9	94.1	87.1	35.5	39.7	A
38	Manama	Bahrain	56.8	70.1	52.6	57.8	41.7	55.5	A
39	Adelaide	Australia	56.3	31.1	91.9	87.1	35.5	36.7	A
40	Toronto	Canada	56.1	28.2	72.2	75.3	58.7	54.1	A

#	City	Country	SLIC	Grw	Via	Cap	Com	Cre	Grd
41	San Juan	Puerto Rico	55.9	48.9	59.2	68.3	61.2	45.8	B
42	Amsterdam	Netherlands	55.7	38.8	75.1	71.8	53.2	42.8	A
43	Tianjin	China	55.5	57	76.7	71.9	26.5	37.1	A
44	Hiroshima	Japan	55.4	45.7	94.6	67.4	24	37.3	A
44	Paris	France	55.4	28.9	78.9	63.9	68.7	45.1	A
46	Gdansk	Poland	55.2	57.3	61.5	65.1	54.9	36.9	A
47	Vancouver	Canada	55	22.3	74	75.3	58.7	54.1	A
48	Bratislava	Slovakia	54.7	54.7	75.3	49.7	56.2	35.3	A
48	Jeju City	South Korea	54.7	35.1	88.7	85.8	25.5	35.6	A
48	Krakow	Poland	54.7	52.3	65.1	65.1	54.9	36.9	A
51	Chiang Mai	Thailand	54.6	61.9	60.7	56.4	73.9	22.8	A
51	Port Louis	Mauritius	54.6	66	76.2	47	25.3	45.4	A
53	Bologna	Italy	54.1	49.2	77.9	66.9	45	29.4	A
54	Kota Kinabalu	Malaysia	53.8	67.3	75.4	52.4	31.4	31.5	A
55	Kuching	Malaysia	53.7	67.3	73.7	52.4	32.6	31.5	A
56	Taichung	Taiwan	53.2	67.8	None	None	0	74.7	Wat chlis t
57	Melaka	Malaysia	53.1	68.2	70.5	52.4	31.7	31.5	A
58	George Town	Malaysia	53	64.7	73.7	52.4	32.6	31.5	A
58	Sapporo	Japan	53	45.7	76.7	67.4	34	37.3	A
60	Salalah	Oman	52.6	69.9	71.1	42	38.5	30.6	A
61	Cordoba	Argentina	52.3	59.5	69.2	78.6	40.6	10.1	A
61	Dubai	United Arab Emirates	52.3	55.8	61.9	62	48.9	31.2	A
61	Guangzhou	China	52.3	41.7	88	71.9	14.5	37.1	A
64	Abu Dhabi	United Arab Emirates	52.2	56.9	60.2	62	48.9	31.2	A
65	Budapest	Hungary	52	41.5	79.3	55.5	59.6	26.2	A
65	Montevideo	Uruguay	52	50.5	61.5	74.6	50.5	24.3	A
67	Hangzhou	China	51.9	42.4	76.7	71.9	27.4	37.1	A
68	Auckland	New Zealand	51.7	20.2	82.6	85.7	40.7	35.1	A
69	Shenzhen	China	51.6	36.5	92.5	71.9	11.5	37.1	A
70	Christchurch	New Zealand	51.4	25.7	80.8	85.7	38.7	30.1	A
71	Muscat	Oman	51.2	64.3	71.1	42	38.5	30.6	A
72	Male	Maldives	50.9	64.4	72.1	39.8	31.2	35.4	A
73	San Jose	Costa Rica	50.8	49.6	56.1	66	46.7	35.8	A
74	Buenos Aires	Argentina	50.3	49.4	71.4	78.6	40.6	10.1	A
74	Chongqing	China	50.3	49.3	61.5	71.9	27.1	37.1	A
76	Hobart	Australia	50.1	29.1	71.6	87.1	33.5	31.7	A
77	Chengdu	China	50	48.2	61.2	71.9	27.3	37.1	A
77	Tbilisi	Georgia	50	59.7	58.9	33.1	None	43.1	Wat chlis t
79	Dunedin	New Zealand	49.4	27.6	73.6	85.7	36.7	27.1	A
79	Kigali	Rwanda	49.4	76.2	68	15.8	50.7	24.8	A

#	City	Country	SLIC	Grw	Via	Cap	Com	Cre	Grd
81	Wellington	New Zealand	49.2	20	77.2	85.7	38.7	30.1	A
82	Casablanca	Morocco	48.8	60	70.3	34.1	26.1	41.2	A
83	Shanghai	China	48.7	33	83.5	71.9	11.5	37.1	A
84	Bucharest	Romania	48.6	50.8	71.7	50.6	26.8	34.8	A
85	Kuala Lumpur	Malaysia	48.5	59.7	68.5	52.4	18.6	31.5	A
86	Helsinki	Finland	48.4	29.8	47.4	81	35.3	52.9	A
86	Kampala	Uganda	48.4	54.6	None	68.7	None	22.4	Wat chlis t
88	Belgrade	Serbia	48.3	51.4	67.5	53.2	33.3	30.3	A
88	Cuenca	Ecuador	48.3	58.6	None	78.8	None	8	Wat chlis t
90	Panama City	Panama	48	59.1	52.7	64.2	31.5	26.9	A
91	Curitiba	Brazil	47.3	59.5	49.9	69.5	22.9	27.5	A
91	Riyadh	Saudi Arabia	47.3	72.3	54.2	78.9	15.4	4.1	A
93	Medellin	Colombia	46.8	55.6	53.2	66.3	35.6	19.6	A
94	Asuncion	Paraguay	46.7	52	None	74.7	None	14.9	Wat chlis t
94	Rabat	Morocco	46.7	61	59.9	34.1	26.1	41.2	A
96	Valencia	Spain	46.5	43.6	100	31.1	12.9	30.4	C
97	Hat Yai	Thailand	46.2	50.2	62.8	56.4	33.9	22.8	A
98	Da Nang	Vietnam	46.1	69.6	None	44.3	43	20.7	C
98	Nizhny Novgorod	Russia	46.1	62.4	62.4	59.2	20.1	15.5	A
100	Phuket	Thailand	46	44.7	68.4	56.4	33.9	22.8	A
101	Florianopolis	Brazil	45.9	58.1	45.1	69.5	22.9	27.5	A
102	Sao Paulo	Brazil	45.8	50.3	53.5	69.5	22.9	27.5	A
103	Amman	Jordan	45.7	48.6	71.6	46.5	8.5	40.8	A
104	Aqaba	Jordan	45.4	52.2	66	46.5	8.5	40.8	A
105	Zagreb	Croatia	44.1	60.5	48.6	17.8	None	42.3	Wat chlis t
106	Phnom Penh	Cambodia	43.1	66.5	None	18.2	None	36.3	Wat chlis t
107	Dar es Salaam	Tanzania	42.8	54.6	None	59.8	None	12.7	Wat chlis t
108	Arequipa	Peru	42.6	67.7	34.1	42.3	37.6	24.7	A
109	Surabaya	Indonesia	42.5	79.4	40.8	35.1	25.6	17.7	A
110	Bogota	Colombia	41.9	50.2	37	66.3	35.6	19.6	A
111	Cebu City	Philippines	41.8	74.2	38.2	40.4	27	17.5	A
112	Nairobi	Kenya	41.6	64.3	69.1	12.5	34.2	14.5	A
113	Pune	India	41.4	58.6	67.7	30.4	23.2	14.6	A
114	Lima	Peru	41	61.3	34.1	42.3	37.6	24.7	A
115	Makati	Philippines	40.8	70.2	38.2	40.4	27	17.5	A
116	Kuwait City	Kuwait	40.7	65	45	65	16.7	1.9	A
117	Moscow	Russia	40.6	40.6	62.4	59.2	20.1	15.5	A
118	Hyderabad	India	40.3	58.7	62.3	30.4	23.2	14.6	A
119	Guadalajara	Mexico	40.2	46.1	40	55.5	47	14.1	A
120	Jakarta	Indonesia	40	74.2	45.3	35.1	10.6	17.7	A

#	City	Country	SLIC	Grw	Via	Cap	Com	Cre	Grd
120	Thimphu	Bhutan	40	60.6	47	25.2	20.1	34.7	A
122	Brno	Czechia	39.8	67.6	38.3	0	None	42.3	Watchlist
123	Accra	Ghana	38.9	74.3	44	14.9	29.6	17.7	A
124	Vilnius	Lithuania	38.7	62.1	7.5	41.6	None	41.4	Watchlist
125	Kumasi	Ghana	38.3	79.1	35.9	14.9	29.6	17.7	A
126	Alexandria	Egypt	38.2	63.7	None	11.3	None	30.6	Watchlist
127	Bengaluru	India	37.9	57.4	53.1	30.4	23.2	14.6	A
128	Ljubljana	Slovenia	37.6	63.1	43.4	0	None	33.2	Watchlist
129	Mexico City	Mexico	37.4	36.4	38.2	55.5	47	14.1	A
129	Windhoek	Namibia	37.4	55.3	30.5	32.1	40.1	25.4	A
131	Chandigarh	India	37.2	59	47.7	30.4	23.9	14.6	A
132	Doha	Qatar	37	44.4	48	54.4	11.1	19.6	A
133	Santo Domingo	Dominican Republic	36.9	31.7	None	53.7	None	28.2	Watchlist
134	Kandy	Sri Lanka	35.9	62.7	40.7	33.7	19.5	11.7	A
135	Cape Town	South Africa	35.6	31.6	44.8	26.3	47.3	29.8	A
135	Riga	Latvia	35.6	62.5	2.3	33.5	None	40.3	Watchlist
137	Chattogram	Bangladesh	35.5	65.2	48.3	10.8	23.3	15.7	A
137	Gaborone	Botswana	35.5	49.7	31.9	22.2	28.6	39	A
139	Colombo	Sri Lanka	35.4	60.7	40.7	33.7	19.5	11.7	A
140	Merida	Mexico	35.2	49.8	13.3	55.5	47	14.1	A
141	Dhaka	Bangladesh	35.1	63.4	48.3	10.8	23.3	15.7	A
142	Johannesburg	South Africa	34.8	31.7	41.2	26.3	47.3	29.8	A
143	Suva	Fiji	34.6	49.6	36.1	47.7	14.8	17.1	A
144	Pokhara	Nepal	31.6	62.5	37.6	12.2	14.8	16.6	A
145	Gothenburg	Sweden	31.5	42.6	7.5	14.5	26.1	63.3	C
146	Kathmandu	Nepal	31.1	60.4	37.6	12.2	14.8	16.6	A
147	Cork	Ireland	30.7	51.4	17.7	13.3	26.9	37.4	C
148	Antwerp	Belgium	29.8	42.6	28	4.4	27.7	40.4	C
149	Dakar	Senegal	28.6	58.2	0.7	10.6	44.6	26.7	A
150	Karachi	Pakistan	27.8	59.3	32.7	8.1	22.8	4.3	A
150	Lahore	Pakistan	27.8	59.5	32.7	8.1	22.8	4.3	A
152	Islamabad	Pakistan	27.6	58.6	32.7	8.1	22.8	4.3	A
153	Munich	Germany	26.9	4.5	48.6	27.1	19.7	36.2	C
154	Bergen	Norway	24.7	44.6	0	3.2	21.7	48.8	C
155	Port Vila	Vanuatu	20.8	22.1	None	26.7	None	13.9	Watchlist
156	Apia	Samoa	16.5	27.9	None	None	None	2.2	Watchlist
157	Port Moresby	Papua New Guinea	5.7	10.3	None	None	None	0	Watchlist

11. Design Properties

Absolute

All thresholds are fixed constants. Adding city #501 does not change scores for cities #1–500. Scores are comparable across editions without recalibration.

Transparent

Every score traces: raw value → piecewise linear mapping → 0–100 normalized score. The complete formula, all anchor tables, and the source code are published.

Penalizes Imbalance

AMPI's variance penalty ensures that cities cannot hide weaknesses behind a single strong dimension. A city scoring 90/20 is penalized more than a city scoring 55/55.

Honest About Gaps

No imputation is performed. Missing indicators are excluded from aggregation and the data gap is disclosed via coverage grades (A, B, C) with corresponding penalties.

Scalable

The same formula works for 50 or 5,000 cities. Fixed benchmarks mean any new city can be scored independently without affecting existing scores.

Anti-Pattern Resistant

Growth does not reward cheapness. Viability measures real disposable income after rent (DI_PPP), not headline GDP. Community penalizes social boredom. Creative penalizes cultural deserts.

12. Glossary

Term	Definition
AMPI	Adjusted Mazziotta–Pareto Index. Aggregation method that penalizes variance across component scores.
Anchor	A fixed (raw value, score) pair used in piecewise linear normalization.
Coverage Grade	Letter grade (A/B/C) indicating how many indicators have data for a given city or pillar.
DI_PPP	Disposable Income at Purchasing Power Parity. Monthly income after tax, rent, utilities, transit, food, adjusted
GPI	Global Peace Index (IEP). Lower values indicate more peaceful countries.
GRI	Government Restrictions Index (Pew Research). Measures government restrictions on religion.
HAQ	Healthcare Access and Quality Index (IHME/Lancet). 0–100 scale.
PISA	Programme for International Student Assessment (OECD). Standardized test scores.
Pillar	One of five thematic dimensions: Growth, Viability, Capability, Community, Creative.
Piecewise Linear	Interpolation method using fixed anchor points to map raw values to 0–100.
PPP	Purchasing Power Parity. Adjusts monetary values for local cost differences.
Provisional	City with Coverage Grade C (≤3 indicators). Carries 15-point penalty.
SLIC Score	The overall city score: AMPI across five weighted pillar scores, minus coverage penalty.

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